The concept of significant digits can be a little confusing, because it has several rules. We are going to try to make the rules as simple as possible so you can remember them. Let's explain how it works through a few problems.

How many significant digits are in the number 8395?

This one is easy by applying rule one which states:

Rule One: Any non-zero number is significant.

So, looking at our number, we see 4 non-zero numbers. The answer, then is four. There are four significant digits in the number 8395. Let's try another one.

How many significant digits are there in the number 8395.5?

Looking at rule one, we can see that there are five non-zero numbers, so our answer is five significant digits. We added a decimal point, which will begin to complicate things in our next question.

How many significant digits are in the number 0.8395?

Looking at rule one again, we see that there are only four non-zero numbers. There is a decimal point. If we don't add another rule to deal with it, we may get confused later. Let's go ahead and add it:

Rule One: Any non-zero number is significant.

Rule Two: Zeroes placed before other numbers are not significant.

Let's try our two rules out again a few times.

How many significant digits are in the number 945687?

According to rule one, there are six significant digits in this number.

How many significant digits are in the number 0.0045?

Rule two provides the key to answering this question. Zeroes placed before other numbers are insignificant. This number has 2 significant digits.

How many significant digits are in the number 8395.005?

According to rules one and two, this number has 5 significant digits. Unfortunately, when we involve zeroes, the problems can get a little more complicated and we need to expand on rule two to deal with them.

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Rule One: Any non-zero number is significant.

Rule Two: Zeroes placed before other numbers are not significant unless they end the number to the right of the decimal point or are between non-zero digits.

Sound confusing? It can be, but let's take another look at our number, 8395.005 with the new rule two in mind. If you guessed 7 significant numbers, you are right. Our new modification to rule two handles this one smoothly. There are four non-zero numbers to the left of the decimal point. Plus one non-zero number to the right of the decimal point, plus the two zeroes to the right of the decimal point that happen to be between two nonzero numbers (the two 5's). That makes seven.

How many significant digits are in the number 8900?

This one is tricky. Because this number ends in two zeroes and the zeroes are to the left of the decimal point, we can't be sure exactly how many significant digits it contains. It has to be at least two, but could possibly be 4 significant digits. In this instance, we assume the smaller number to be correct, which is two significant figures. If you wanted to write a number such as this that ended in zeroes before the decimal point and you intended for all the digits to be significant, you should write the number using scientific notation such as 8.900 X 103. That way people would know that because the zeroes occur after the decimal point, they are significant. If it was written 8.9 X 103, the number would still be 8900, but it would only be 2 significant digits.

How many significant digits are in the number 90.4050?

Again, you'll have to look closely at rule two to answer this one. For starters, the answer is 6, but let's see how we came up with the answer. There are three non-zero digits in the number (9, 4, and 5) and according to rule one, they MUST be significant. Now look at the two zeroes to the right of the decimal point. Rule two clearly states that zeroes to the right of the decimal point are significant, so that makes a total of 5 significant digits. But what about the zero to the right of the nine? Rule two also states that a zero is significant if it appears between two other digits or at the end of the number if it is to the right of the decimal point. That gives us a total of 6 significant digits.

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Significant Digits Practice Problems

How many significant digits are in each of the following numbers?

1. 8200
2. 8.2×10^3
3. 4178
4. 203
5. 0.0097
6. 9.7×10^{-3}
7. 1.0×10^1
8. 410.13
9. 16.0390
10. 5.9×10^6
11. 1.92×10^3
12. 3.14159
13. 0.1
14. 25
15. 2500
16. 2500.003
17. 19.9
18. 1.99
19. 900
20. 9.00
21. 8627.104
22. 50
23. 49
24. 1000
25. 1.000
26. 1×10^0
27. 10943.06
28. 612.0
29. 5.03
30. 1900
31. 4.3×10^3
32. 16
33. 21030
34. 21030.5
35. 0.79
36. 1916
37. 6.4×10^{14}
38. 0.00005101
39. 0.0030
40. 10
41. 100
42. 100.01
43. 793.601
44. 2.0×10^3
45. 145
46. 0.212
47. 1090
48. 64
49. 1031.0031
50. 0.101010

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Significant Digits Answers

1. 2
2. 2
3. 4
4. 3
5. 2
6. 2
7. 2
8. 5
9. 6
10. 2
11. 3
12. 6
13. 1
14. 2
15. 2
16. 7
17. 3
18. 3
19. 1
20. 3
21. 7
22. 1
23. 2
24. 1
25. 4
26. 1
27. 7
28. 4
29. 3
30. 2
31. 2
32. 2
33. 4
34. 6
35. 2
36. 4
37. 2
38. 4
39. 2
40. 1
41. 1
42. 5
43. 6
44. 2
45. 3
46. 3
47. 3
48. 2
49. 8
50. 6

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